

CO1-AT-1-Short Answer Questions

DSA0404 – Fundamentals of Data Science

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1. What is Data Science? Explain its primary objective.

Data Science is the field of study that combines statistics, programming, and machine learning to extract useful information from data. It helps in analyzing large amounts of structured and unstructured data. The primary objective of Data Science is to discover meaningful patterns and insights. These insights support better decision-making and solve real-world problems.

2. Why is Data Science important in today's digital world? Mention any four reasons.

Data Science is important because it helps organizations make data-driven decisions. It improves business efficiency by identifying trends and patterns. It enables accurate predictions using historical data. It also supports automation, personalization, and innovation across various industries.

3. List any four benefits of Data Science in business and society.

- Improves decision-making using data insights.
- Increases business efficiency and productivity.
- Helps detect fraud and manage risks.
- Enhances healthcare, education, and public services through better analysis.

4. Mention four real-world applications of Data Science.

- Healthcare: Disease prediction and medical diagnosis.
- Finance: Fraud detection and credit scoring.
- E-commerce: Product recommendations and customer analysis.
- Transportation: Traffic prediction and route optimization.

5. What are the different facets of data? Briefly explain each.

The main facets of data are structured, semi-structured, unstructured, and metadata. Structured data is organized in rows and columns. Semi-structured data has some organization but not a fixed format. Unstructured data includes text, images, videos, and audio. Metadata describes information about other data.

6. Differentiate between structured, semi-structured, and unstructured data with suitable examples.

Structured Data	Semi-Structured Data	Unstructured Data
Organized in tables	Partially organized	No predefined structure
Easy to query	Flexible format	Difficult to process
Stored in databases	Stored in XML/JSON	Stored as text, images, videos
Example: Student database	Example: XML, JSON	Example: Emails, Photos, Videos

7. What is Big Data? State its key characteristics (5 Vs).
Big Data refers to extremely large and complex datasets that cannot be processed using traditional methods. It is characterized by the **5 Vs**:

- **Volume** – Large amount of data.
- **Velocity** – Speed of data generation.
- **Variety** – Different types of data.
- **Veracity** – Accuracy and reliability of data.
- **Value** – Useful insights obtained from data.

8. Explain the Big Data ecosystem. Name any four technologies used in it.
The Big Data ecosystem consists of tools and technologies used for storing, processing, analyzing, and visualizing large datasets. It supports handling massive amounts of structured and unstructured data. Common technologies include **Hadoop**, **Spark**, **Hive**, and **HBase**. These technologies improve scalability and processing speed.

9. What are the major phases of the Data Science process?
The major phases are:

1. Data Collection
2. Data Cleaning
3. Data Integration
4. Data Transformation
5. Data Analysis
6. Model Building
7. Model Evaluation
8. Deployment and Presentation of Results

10. What is data retrieval? Mention two common sources from which data can be retrieved.

Data retrieval is the process of collecting or accessing data from different storage systems for analysis. It ensures the required data is available for processing. Two common sources are Databases (SQL) and Web APIs. Data can also be retrieved from files such as CSV or Excel.

11. Why is data collection considered an important step in the Data Science process?

Data collection is important because the quality of the final analysis depends on the collected data. Accurate and relevant data leads to better predictions and decisions. Poor-quality data can produce incorrect results. Therefore, collecting reliable data is the foundation of Data Science.

12. What is data cleansing? List any four common data-cleaning techniques.

Data cleansing is the process of identifying and correcting errors in a dataset. It improves data quality before analysis. Common techniques include:

- Removing duplicate records.
- Handling missing values.
- Correcting inconsistent data.
- Removing outliers and invalid entries.

13. What is data integration? Why is it necessary in Data Science?

Data integration is the process of combining data from multiple sources into a single dataset. It provides a unified view of information. Integration improves data consistency and completeness. It enables better analysis and more accurate decision-making.

14. Explain data transformation with suitable examples.

Data transformation is the process of converting data into a suitable format for analysis. It includes normalization, aggregation, encoding, and scaling. For example, converting dates into a standard format or converting categorical values such as "Male" and "Female" into numerical values. It improves the performance of machine learning models.

15. What is data analysis? Mention any four techniques used for data analysis.

Data analysis is the process of examining data to discover useful information and patterns. It helps in making informed decisions. Four common techniques are:

- Statistical Analysis
- Regression Analysis
- Classification
- Clustering

16. Differentiate between descriptive, predictive, and prescriptive analytics.

Descriptive Analytics	Predictive Analytics	Prescriptive Analytics
Explains past events	Predicts future outcomes	Suggests the best action
Uses historical data	Uses machine learning	Uses optimization techniques
Answers "What happened?"	Answers "What will happen?"	Answers "What should be done?"
Example: Sales report	Example: Sales forecasting	Example: Route optimization

17. What is meant by building a model in Data Science? Mention the basic steps involved.

Building a model means creating a mathematical or machine learning algorithm that learns patterns from data to make predictions. The basic steps include selecting features, choosing an algorithm, training the model, testing it, and tuning its parameters. A well-trained model provides accurate predictions for new data.

18. Why is model evaluation important? Name any four evaluation metrics used for classification models.

Model evaluation measures how well a machine learning model performs on unseen data. It helps compare models and identify errors before deployment. Common evaluation metrics are:

- Accuracy
- Precision
- Recall
- F1-Score

19. Why is presenting findings an essential part of the Data Science process? Mention any four visualization tools or techniques.

Presenting findings helps stakeholders understand the insights obtained from data analysis. It supports effective communication and better decision-making. Common visualization tools and techniques include: Tableau, Power BI, Matplotlib, Bar Charts and Pie Charts

20. How are Data Science models deployed into real-world applications? Give any four examples of Data Science applications.

After evaluation, Data Science models are deployed into real-world systems using web applications, mobile apps, cloud services, or APIs. Deployment allows users to make predictions using live data. Examples include:

- Spam email detection
- Loan approval systems
- Recommendation systems (Netflix, Amazon)
- Disease prediction in healthcare